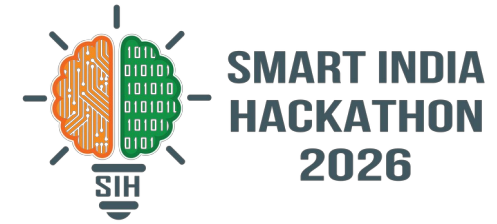


SMART INDIA HACKATHON 2026



TOURINSIGHT

Smart, Safe & Personalized Travel Companion

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SMART INDIA HACKATHON 2026

Problem Statement ID: SIH26204

Problem Statement Title: A solution/idea that can boost the current situation of the tourism industries including hotels, travel and others.

Theme: Travel & Tourism

PS Category: Software

Team ID:130384

Team Name:TourInsight



IDEA TITLE

TourInsight - Context-Aware AI Travel Ecosystem

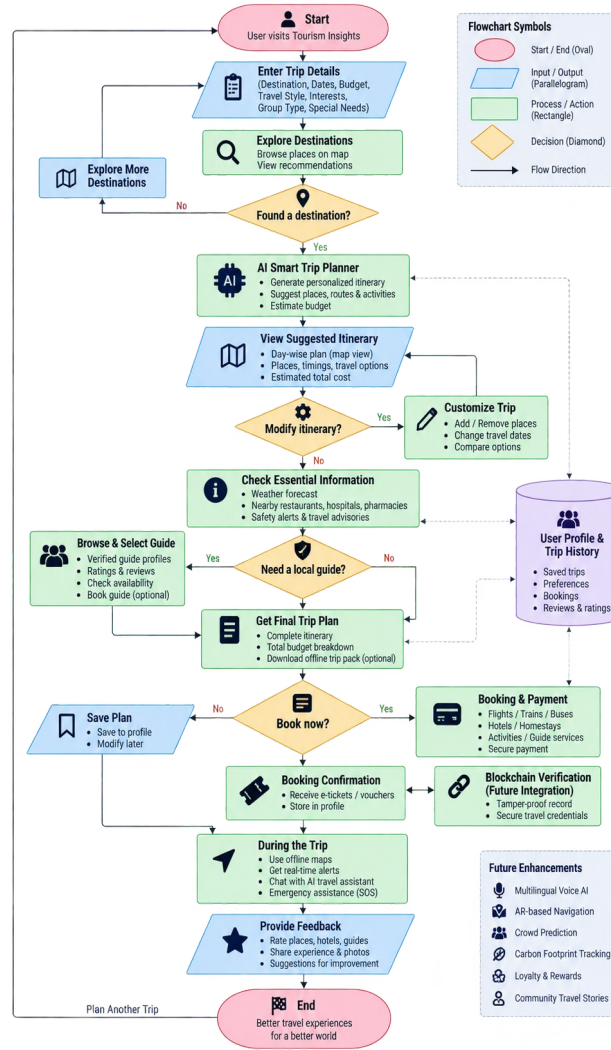
Proposed Solution

Current tourism planning is fragmented because travellers use different platforms for maps, weather, hotels, flights, buses, cabs, food, guides, hospitals and safety. Even after giving the same destination, date, budget and preferences repeatedly, most systems still return generic recommendations.

Following is our **Idea / Approach**:

- **Context-Aware Trip Planner**: captures starting location, destination, start date, transport, days, pace, interests, style, budget, travellers, women/girls in group and guide need.
- **Interactive World Explorer**: users select countries visually and move naturally from exploration to planning.
- **AI Itinerary Engine**: generates day-wise journeys based on pace, interests, budget and travel style instead of only listing famous places.
- **Tourist Guide Trust Layer**: structured profiles show verification, languages, experience, specialization, ratings, availability and transparent pricing.
- **Safety + Essentials Layer**: restaurants, hospitals, police stations, pharmacies, weather and emergency data are connected to the journey.
- **Future Booking Ecosystem**: flights, hotels, cabs, buses, guides and activities will be bookable from the same travel context.
- **Blockchain Trust Layer**: digital tickets and booking confirmations can be verified using tamper-resistant records.
- **Feedback + Agile Improvement**: every traveller rating becomes an input for future releases.

User Flow Diagram



Uniqueness

- **Traveller Context Engine** - one input profile reused across the journey, not repeated in every service.
- **AI Itinerary Personalization** - plans change based on budget, pace, interests, travel style and group type.
- **Interactive Maps** - country selection, route understanding and nearby discovery make the plan visual.
- **Tourist Guide Transparency** - verification, languages, experience, reviews and price build trust.
- **Blockchain Booking Vision** - future ticket layer for tamper-resistant digital tickets and traceable transactions.
- **Agile Feedback Loop** - product evolves from real customer ratings and market situations.

How It Addresses the Problem

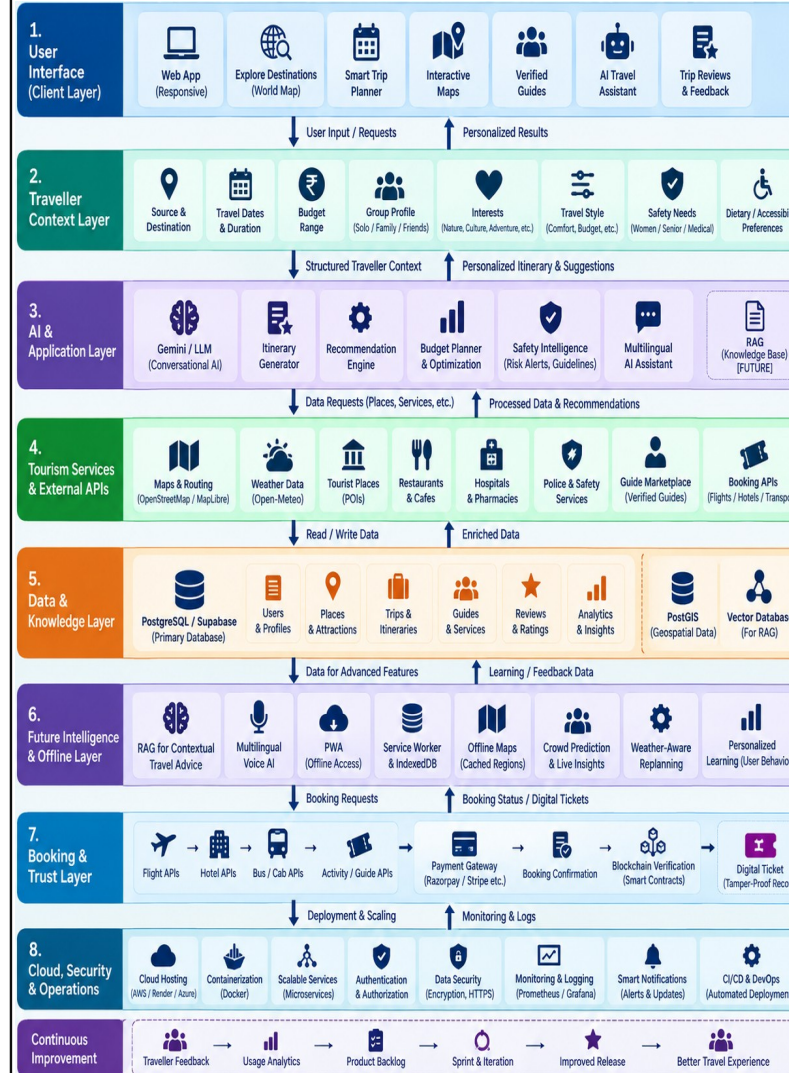
- **Fragmented Travel Apps?** → One continuous platform connecting planning, maps, safety, guides and assistance.
- **Generic Plans?** → AI uses interests, pace, style, budget and traveller profile to generate a contextual trip.
- **Safety Gaps?** → Essentials layer brings hospitals, police, pharmacies, SOS roadmap and women-safety context closer to the journey.
- **Guide Trust Issues?** → Structured guide profiles support better comparison and transparent pricing.
- **Market Change?** → Feedback analytics + Agile software management allow continuous improvement instead of a fixed product.
- **Booking Trust?** → Future payments and blockchain verification reduce fake or modified ticket records.

Tech stack



- **Frontend / UI:** responsive web interface, interactive dashboard, smart forms, website-based prototype and map-first destination exploration.
- **AI Layer:** Gemini / LLM-driven itinerary generation, preference understanding, budget-aware suggestions and contextual travel chatbot.
- **Data Models:** Trip, TripStop, Place, Guide, GuideReview, Feedback and OfflinePack keep the product structured instead of static.
- **Map Layer:** interactive world map, country selection, destination visualization, markers, coordinates and future route intelligence.
- **Safety Data:** hospitals, police, pharmacies, women-safety context, SOS roadmap and real-time disaster/weather alert plan.
- **Booking APIs:** future flight, hotel, cab, bus, guide and activity booking with one-way, round-trip and multi-city support.
- **Blockchain Layer:** future tamper-resistant ticket hash, booking audit trail, transaction traceability and fraud-reduction mechanism.
- **Advanced AI:** RAG + vector database for trusted tourism knowledge; multilingual voice assistant for Indian and global travellers.
- **Geospatial DB:** PostgreSQL + PostGIS for nearby search, distance calculation, crowd intelligence and location-based recommendations.
- **Offline Tech:** PWA, Service Worker, IndexedDB and offline trip pack to support low-network travel situations.

Architecture diagram



AI Assistant + Smart Modules

- **Smart Trip Planner:** converts natural travel requirements into itinerary logic using destination, date, transport, pace, interests, travel style and budget.
- **AI Travel Chatbot:** answers contextual questions such as “What should I visit this evening?”, “Suggest a cheaper option”, or “What is near me?”.
- **Safety Intelligence:** expands from hospitals and police to SOS live-location sharing, offline emergency card, women helplines and real-time weather/disaster alerts.
- **Personalization Engine:** future dietary needs, accessibility needs, kids/elderly travel, and solo-female traveller mode.
- **Travel Utility Layer:** future currency converter, phrasebook, packing checklist, calendar export and downloadable PDF trip summary.
- **Sustainability Layer:** carbon footprint comparison and highlighting locally-owned stays/eateries.



Technical Feasibility

- **AI Planner Feasibility:** LLMs can interpret traveller preferences, trip duration, budget and interests to generate personalized itinerary drafts.
- **Map Feasibility:** interactive maps and geospatial coordinates support country exploration, destination discovery, route understanding and nearby essential-service search.
- **Data Feasibility:** the current MVP already separates trips, stops, places, guides, reviews, feedback and offline-pack structures.
- **Blockchain Feasibility:** planned only for booking trust - ticket verification, immutable confirmation hash and transaction audit trail.
- **Offline Feasibility:** PWA + Service Worker + IndexedDB can preserve critical trip data for low-network areas.
- **AI Upgrade Feasibility:** RAG and vector search can ground chatbot answers in trusted tourism data.
- **Scalable:** new cities, languages, guide categories and booking APIs can be added without redesigning the core.



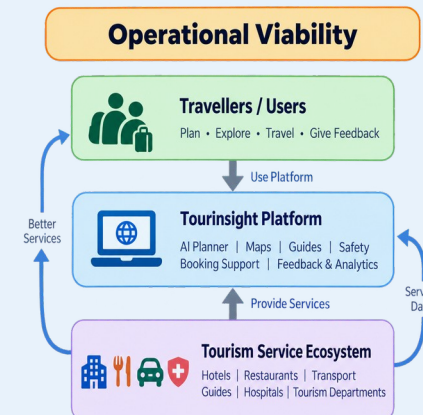
Feasibility Flow



Operational Viability

- **Traveller-Friendly:** one journey context reduces repeated searching across multiple disconnected apps.
- **Market-Ready MVP:** the current prototype demonstrates world map exploration, smart planner, generated itinerary, guides, essentials, chatbot and feedback.
- **Service Partnerships:** hotels, cab operators, bus/flight aggregators and local guides can be onboarded phase by phase.
- **Customer Feedback:** feedback is treated as a development input, not just a rating screen.
- **Financial Viability:** commissions, guide booking fees, premium AI planning, affiliate partnerships and promoted local services can sustain the model.
- **Social Viability:** local guides, small restaurants, stays and lesser-known destinations gain visibility.
- **Challenge Strategy:** offline maps, real-time APIs, payment integration and blockchain verification will be introduced in controlled phases after MVP validation.

Operational Viability



Impacts

- For Travellers: Personalized planning, budget awareness, guide transparency, AI assistance and faster access to essentials reduce stress before and during travel.
 - For Local Guides: Verified profiles, ratings, availability and transparent price display create digital visibility, trust and earning opportunities.
 - For Local Businesses: Restaurants, hotels, cabs, shops and activity providers can be discovered through contextual recommendations.
 - For Safety: SOS roadmap, emergency card, women helplines, hospitals and police discovery create a stronger safety-first tourism layer.
 - For Economy: booking commissions, local tourism discovery and wider destination promotion can support jobs and tourism revenue.
 - For Accessibility: dietary needs, wheelchair-friendly routes, kids/elderly mode and solo-female mode make planning more inclusive.
 - For Sustainability: carbon footprint comparison and locally-owned stay/eatery highlights encourage responsible travel.
 - For Governance: feedback analytics can reveal service gaps, popular routes, crowd patterns and improvement priorities.
- Potential impact: TourInsight can act as a digital bridge between travellers, local service providers, guides and future tourism decision-makers.

Future scope

- **Unified Booking:** flights, hotels, cabs, buses, guides and activities inside the TourInsight journey.
- **Journey Types:** One Way, Round Trip and Multi-City planning/booking without recreating separate trips.
- **Cabin Experience:** Economy, Premium Economy, Business Class and First Class with comfort, baggage, meals, lounges and service preferences.
- **Blockchain Tickets:** tamper-resistant digital tickets, traceable transactions and booking verification layer.
- **Safety Expansion:** SOS live location, offline emergency card, region-specific women helplines and disaster alerts tied to the current stop.
- **Personalization Expansion:** dietary needs, accessibility needs, travelling with kids/elderly and solo-female traveller mode.
- **Group Features:** shared itinerary, expense splitting and in-app voting for group decisions.
- **Travel Utilities:** currency converter, phrasebook, packing checklist, calendar export and downloadable PDF trip summary.
- **AI Expansion:** RAG, vector database, multilingual voice assistant, weather-aware replanning and crowd prediction.

Advantages



Benefits

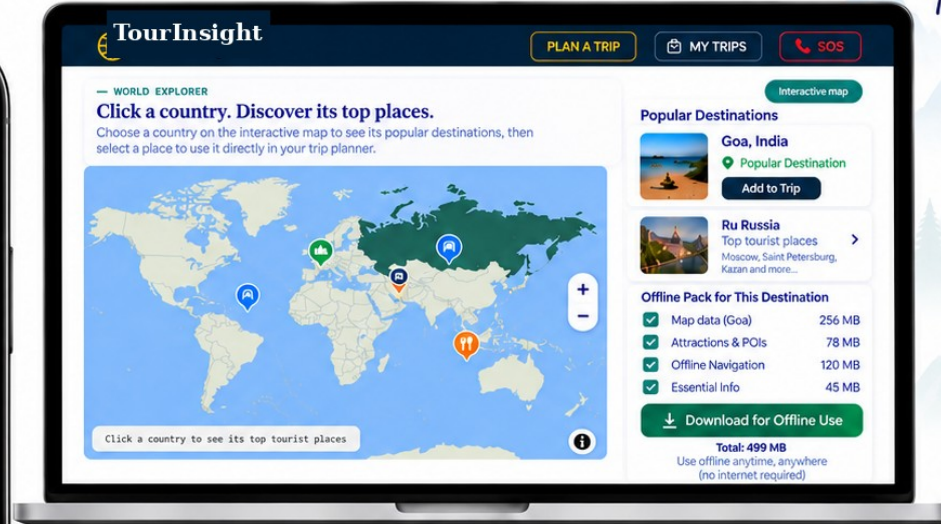


Business model

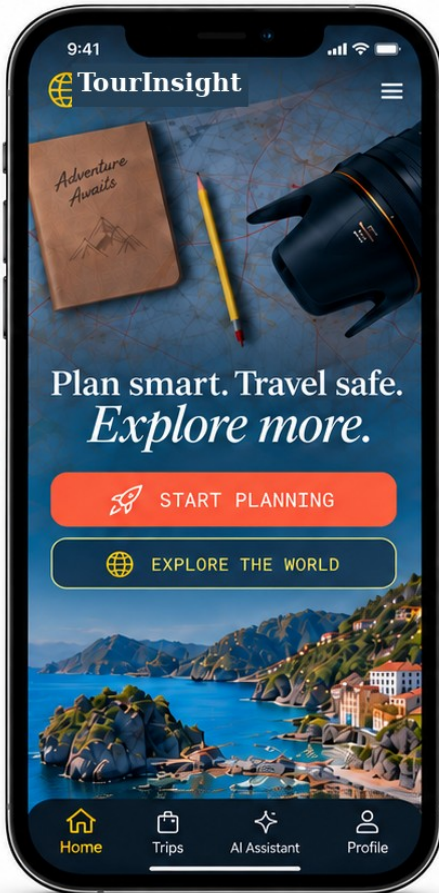
- **Target Customers:** domestic tourists, international travellers, families, solo travellers, student groups, local guides, hotels, restaurants, cab operators, bus operators, tourism service providers and destination partners.
- Revenue Channels:
 - **Booking commissions** from flights, hotels, cabs, buses, activities and local guides.
 - **Premium AI planning tier** for detailed itineraries, PDF summaries, advanced personalization and group collaboration.
 - **Sponsored local-business** visibility with transparent listing policies.
 - **B2B partnerships** with tourism boards, travel operators, hotels and hospitality partners.
 - Future **blockchain** verification service for secure ticket validation and audit records.
- Operational Rollout: Start with high-tourism cities, validate with real users, onboard guides and local businesses, integrate booking APIs, then expand state-wise and nationally.
- **Scalability:** Selected cities → state tourism circuits → Pan-India → multilingual support → international destinations.



Sample UI developed by the team



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Plan
Travel
Together



Reference Article & Sources

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- **Blockchain in Tourism** — <https://link.springer.com/article/10.1007/s40558-022-00229-6>
- **India Tourism Data** — <https://data.tourism.gov.in/tourismdata>
- **Verified Tourist Guides** — <https://iitf.gov.in/>
- **Emergency / SOS Support** — <https://www.mha.gov.in/en/commocontent/emergency-response-support-system-erss>
- **Weather & Risk Alerts** — <https://mausam.imd.gov.in/>

Prototype Links

- **Live Website:** <https://skilled-tour-insight-go.base44.app/>



Scan for Prototype



Integrated Maps
Explore destinations with
real-time and offline maps



Offline Access
Download trip packs and
use without internet



Smart Trip Planner
Plan personalized trips
with AI-powered suggestions



AI Travel Assistant
Get instant help,
even offline



Blockchain Booking
Secure & transparent
bookings with blockchain
technology



Travel Safer
Verified information
and emergency support